

Topic 2 Summary: Scalar and Vector Quantities

- Definitions: scalar, vector
- Geometrical representation of vectors
- Coordinate (component) representation of vectors
- Vector algebra: addition, subtraction and scalar multiplication of vectors in both geometrical and component form.
- The unit vector, including standard basis vectors
- Magnitude of a vector
- Composition and resolution of vectors
- Differentiation and integration of vector functions

SCALAR & VECTOR QUANTITIES.

Certain physical quantities such as mass or the absolute temperature at some point only have magnitude. These quantities can be represented by numbers alone, with the appropriate units, and they are called SCALARS.

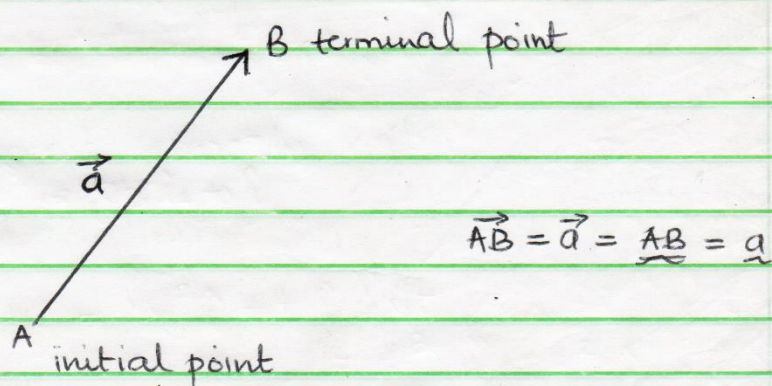
There are, however, other physical quantities which have both magnitude and direction. Examples are velocity, acceleration, force, momentum etc. These physical quantities are represented mathematically by VECTORS.

Definitions

- 1) A scalar is a quantity which has only magnitude
- 2) A vector is a quantity that has both magnitude and direction.

GEOMETRICAL REPRESENTATION OF VECTORS

Vectors are represented by directed line segments where the length of the line is the magnitude and the direction of the line is the direction of the vector.



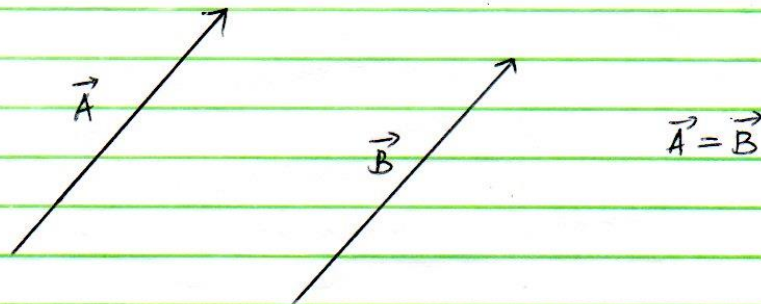
The magnitude of a vector $\vec{A}$ is denoted as $|\vec{A}|$.

Free Vectors : are those whose value is not affected by change in position.

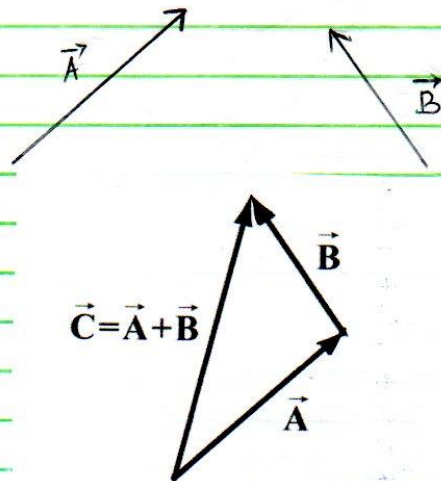
Localised Vectors : are affected in value by change in position e.g. Force, because a change in line of action of a force alters its effect on a body and thus its value.

VECTOR ALGEBRA.

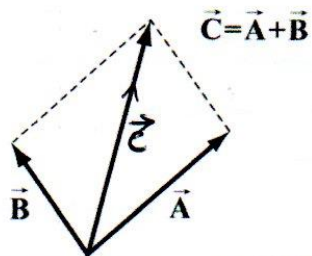
- 1) Two vectors $\vec{A}$ and $\vec{B}$ are equal if and only if they have the same magnitude and direction regardless of their initial position.



- 2) Vector Addition (Finding the resultant of two or more vectors).



Triangle law of addition

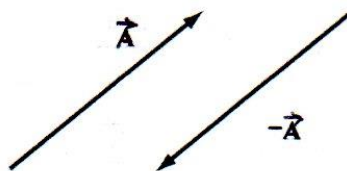


Parallelogram law of addition

(place vectors tail to tail).

$$\vec{C} = \vec{A} + \vec{B} = \vec{B} + \vec{A}$$

3) Negative Vector

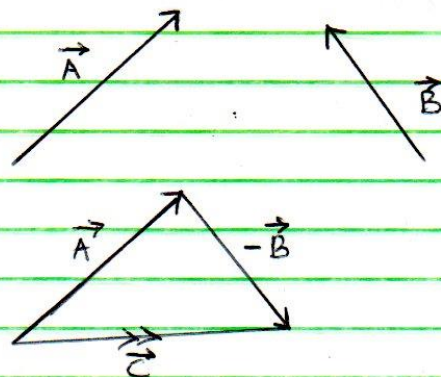


A vector opposite in direction to vector $\vec{A}$ is the negative of $\vec{A}$ denoted $-\vec{A}$.

$$\vec{A} + (-\vec{A}) = \vec{0} \text{ (zero or null vector)}$$

The null (zero) vector has zero magnitude and no specific direction.

4) Subtraction of Vectors

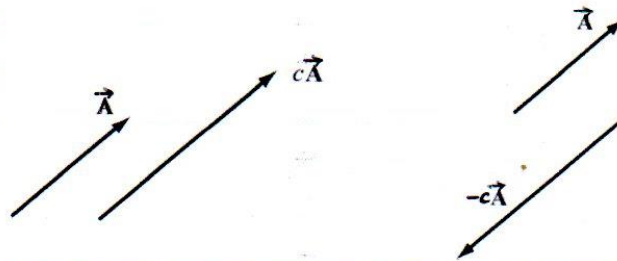


$$\vec{A} - \vec{B} = \vec{A} + (-\vec{B})$$

5) Scalar Multiplication of Vectors.

Vectors can be multiplied by real numbers (Scalars).
The product of a vector $\vec{A}$ by a scalar c is $c\vec{A}$.

If $c > 0$, then $c\vec{A}$ and $\vec{A}$ are in the same direction
however if $c < 0$, then $c\vec{A}$ and $\vec{A}$ are in opposite
directions.

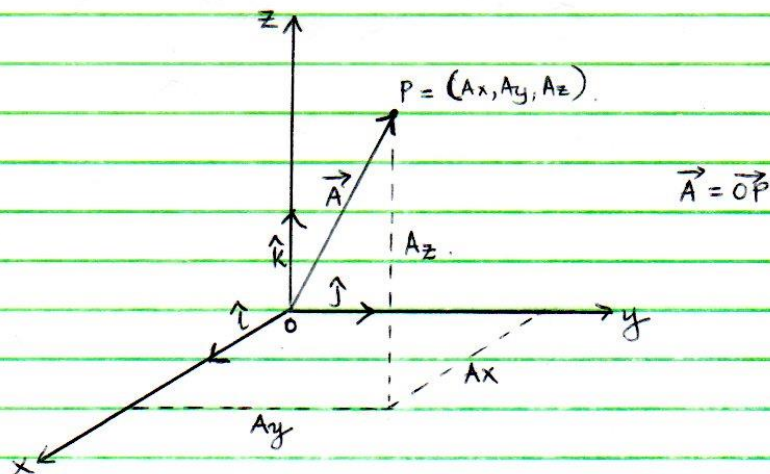


The Unit Vector

- A unit vector is a vector with a magnitude of 1 unit.
- If $\vec{A}$ is a vector with magnitude $|\vec{A}| \neq 0$, then a unit vector in the direction of $\vec{A}$ is $\frac{\vec{A}}{|\vec{A}|}$

ie $\hat{u} = \frac{\vec{A}}{|\vec{A}|}$

B COORDINATE REPRESENTATION OF VECTORS



Unit vectors in the directions x, y, z in the Cartesian coordinate system above are represented by vectors $\hat{i}, \hat{j}$ and $\hat{k}$ respectively, where

$$\hat{i} = \langle 1, 0, 0 \rangle$$

$$\hat{j} = \langle 0, 1, 0 \rangle$$

$$\hat{k} = \langle 0, 0, 1 \rangle$$

$$\text{Then } \vec{OP} = \vec{A} = A_x \hat{i} + A_y \hat{j} + A_z \hat{k}$$

$\vec{OP}$ = Position Vector of point P.

$$\begin{aligned} \vec{A} \text{ can be denoted: } \vec{A} &= \langle A_x, A_y, A_z \rangle \\ &= \begin{pmatrix} A_x \\ A_y \\ A_z \end{pmatrix} = \begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix} \end{aligned}$$

The addition, subtraction and scalar multiplication of vectors in coordinate form is as follows:

$$\begin{aligned} \text{Let } \vec{A} &= A_1 \hat{i} + A_2 \hat{j} + A_3 \hat{k} \\ \vec{B} &= B_1 \hat{i} + B_2 \hat{j} + B_3 \hat{k}, \text{ then} \end{aligned}$$

$$\vec{A} \pm \vec{B} = \langle A_1 \pm B_1, A_2 \pm B_2, A_3 \pm B_3 \rangle$$

If c is a scalar, then

$$\begin{aligned} c\vec{A} &= cA_1 \hat{i} + cA_2 \hat{j} + cA_3 \hat{k} \\ &= \langle cA_1, cA_2, cA_3 \rangle \end{aligned}$$

Magnitude of a Vector

If $\vec{A} = A_1\hat{i} + A_2\hat{j} + A_3\hat{k}$, then

$$|\vec{A}| = \sqrt{A_1^2 + A_2^2 + A_3^2}$$

Laws of Vector Algebra.

If $\vec{A}$, $\vec{B}$ and $\vec{C}$ are vectors and p & q are scalars, then

1. $\vec{A} + \vec{B} = \vec{B} + \vec{A}$
2. $\vec{A} + (\vec{B} + \vec{C}) = (\vec{A} + \vec{B}) + \vec{C}$
3. $\vec{A} + \vec{0} = \vec{A}$
4. $\vec{A} + (-\vec{A}) = \vec{0}$
5. $p(\vec{A} + \vec{B}) = p\vec{A} + p\vec{B}$
6. $p(q\vec{A}) = q(p\vec{A}) = pq(\vec{A})$
7. $(p+q)\vec{A} = p\vec{A} + q\vec{A}$
8. $1\vec{A} = \vec{A}$

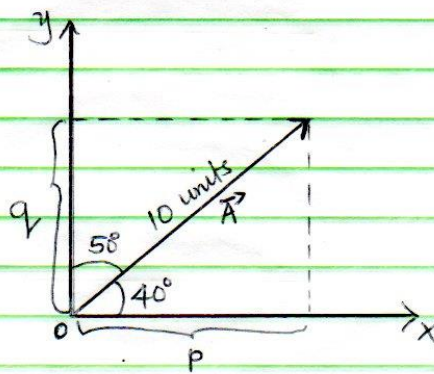
Examples (In class)

RESOLUTION OF VECTORS

A vector can be expressed as two separate vectors in any two directions. This is called resolving a vector into two components.

Normally, vectors are resolved into rectangular components i.e. components that are at right angles to each other. This is mainly because the values of these components are easily calculable by trigonometry.

For example:- If we wish to resolve a vector of magnitude 10 units into two components making angles of 40° and 50° with it as shown, we get the following:

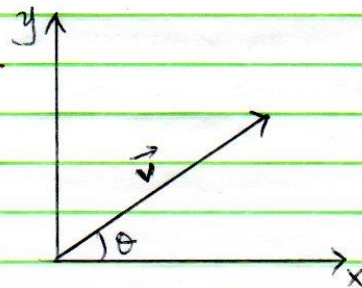


$$\cos 40^\circ = \frac{p}{10} \Rightarrow p = 10 \cos 40^\circ$$

$$\sin 40^\circ = \frac{q}{10} \Rightarrow q = 10 \sin 40^\circ$$

$$\therefore \vec{A} = p \hat{i} + q \hat{j} = 10 \cos 40^\circ \hat{i} + 10 \sin 40^\circ \hat{j}$$

In general, if a vector $\vec{v}$ lies in the xy-plane, is of magnitude $|\vec{v}|$ and makes an angle θ with the x-axis as shown below:



Then $\vec{v} = |\vec{v}| \cos \theta \hat{i} + |\vec{v}| \sin \theta \hat{j}$

COMPOSITION OF VECTORS

The composition of two or more vectors in the xy-plane e.g. $\vec{r}_1, \vec{r}_2, \vec{r}_3, \dots$ where

$$\vec{r}_1 = x_1 \hat{i} + y_1 \hat{j}$$

$$\vec{r}_2 = x_2 \hat{i} + y_2 \hat{j}$$

$$\vec{r}_3 = x_3 \hat{i} + y_3 \hat{j} \dots \text{etc}$$

is given by the sum

$$\vec{R} = (x_1 + x_2 + x_3 + \dots) \hat{i} + (y_1 + y_2 + y_3 + \dots) \hat{j}$$

$$\text{and } |R| = \sqrt{(x_1 + x_2 + x_3 + \dots)^2 + (y_1 + y_2 + y_3 + \dots)^2}$$

$$\text{while } \tan \theta = \frac{(y_1 + y_2 + y_3 + \dots)}{(x_1 + x_2 + x_3 + \dots)}$$

Examples (in class)

Vectors can be differentiated/Integrated by considering each component separately e.g. if

$$\vec{a} = t^2 \hat{i} + t \hat{j}$$

$$\text{then } \frac{d\vec{a}}{dt} = 2t \hat{i} + \hat{j}$$

$$\text{and } \int \vec{a} dt = \int (t^2 \hat{i} + t \hat{j}) dt = \frac{t^3}{3} \hat{i} + \frac{t^2}{2} \hat{j} + \vec{C}$$

where $\vec{C}$ is the ^{vector} constant of integration.